

Line Segments Trace I

Time limit: 1.00 s

Memory limit: 512 MB

There are n line segments whose endpoints have integer coordinates. The left x-coordinate of each segment is 0 and the right x-coordinate is m . The slope of each segment is an integer.

For each x-coordinate $0, 1, \dots, m$, find the maximum point in any line segment.

Input

The first line has two integers n and m : the number of line segments and the maximum x-coordinate.

The next n lines describe the line segments. Each line has two integers y_1 and y_2 : there is a line segment between points $(0, y_1)$ and (m, y_2) .

Output

Print $m + 1$ integers: the maximum points for $x = 0, 1, \dots, m$.

Constraints

- $1 \leq n, m \leq 10^5$
- $0 \leq y_1, y_2 \leq 10^9$

Example

Input	Output
4 5 1 6 7 2 5 5 10 0	10 8 6 5 5 6